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Electronic transport and noise characterization in MoS_2

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Abstract

A study of the diffusivity in Molybdenum disulfide as a function of the applied electric field and temperature has been developed in this work by means of an ensemble Monte Carlo simulator. The analysis of the different scattering types indicates that the transitions to the upper valleys have an important effect over the noise and the velocity fluctuations, showing an influence over the frequency response of the diffusivity. Indeed, the low-frequency value of the diffusion coefficient decreases with the increase of the applied electric field, presenting a wider white-noise regime. The values of the diffusion coefficient are one order of magnitude lower than in Silicon and two orders than in graphene. Our results also evidence that the ambient temperature is a key factor to obtain competitive mobilities in this material: at room temperature the degradation of transport produced by the strong scattering activity in the upper valleys leads to mobilities close to 150 $cm^2 V^{-1} s^{-1}$, while in the lower valley-dominated regime at reduced temperatures the mobility rises up to 5400 $cm^2 V^{-1} s^{-1}$.

Keywords: ensemble Monte Carlo, metal dichalcogenides, diffusive transport, Molibdenum disulfide, electronic noise

(Some figures may appear in colour only in the online journal)

1. Introduction

The interest in two dimensional (2D) metal dichalcogenides has been boosted in the last years [1, 2] due to their inherent band gap and their electrical and optical features, among other properties. Molybdenum disulfide (MoS_2) is one of the most renowned semiconductors of this type because of its gap duality to a certain extent, being indirect for bulk material and direct for the monolayer MoS_2, with a value of 1.8 eV in this case [3]. Different electrical properties of its monolayer version have been studied, such as its conductivity and mobility [4, 5]. Besides, the performance of prototype MoS_2-FETs has already been tested [5, 6] opening new encouraging perspectives for ultra-scaled devices. Indeed, different experiments have been developed to determine the field effect mobility in several MoS_2 devices, such as MoS_2 TFTs [7], monolayer transistors [8], back-gated MoS_2 FETs [9] or MoS_2 FETs on polymethyl methacrylate [10]. In this context, in order to elucidate the potential of this material it becomes mandatory a thorough analysis of electronic transport, where advanced numerical models can play an important role. In the case of the Monte Carlo technique [11], its stochastic treatment of charge transport processes makes it highly suitable for this task. In the literature some Monte Carlo models have already explored the basic properties of this material: Li *et al* presented an analysis of the velocity-electric field curves by means of their full-band Monte Carlo simulator, focusing on the mobility and the estimated saturation velocity of MoS_2 [12]. Ferry used an Ensemble Monte Carlo simulator for discussing mainly about the electron velocity, the mobility and some scattering processes in MoS_2 [13]. However, a study about velocity fluctuations and their relation with the diffusion coefficient has never been performed so far. This is important, since velocity fluctuations are related to the origin of the microscopic thermal noise source, providing important information for analysing the high-frequency behaviour of the material. Besides, this study also provides an alternative microscopic based way of obtaining the electron mobility through the analysis of the diffusive properties of electronic transport.

In this paper we will focus on a Monte Carlo study of the carrier velocity fluctuations in MoS_2. We will firstly analyse

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the dependence of the drift velocity with the electric field. After that, the study of the velocity fluctuations of carriers for different electric field and temperature conditions will be performed. The range of electric field attained is justified by simulations [12, 13] and experiments in monolayer MoS_2 [6, 14]. Nanosecond-duration electrical pulses can be used to perform such high field meaurements in order to control heating effects in 2D materials. The autocorrelation function of the velocity fluctuations will be analysed with particular attention as well as the scattering mechanisms events suffered by the particles and the upper valley occupation. Through this analysis, the sources of noise in MoS_2 will be properly characterized. Finally, the stress is also put over determining the diffusion coefficient and the carrier mobility of this material.

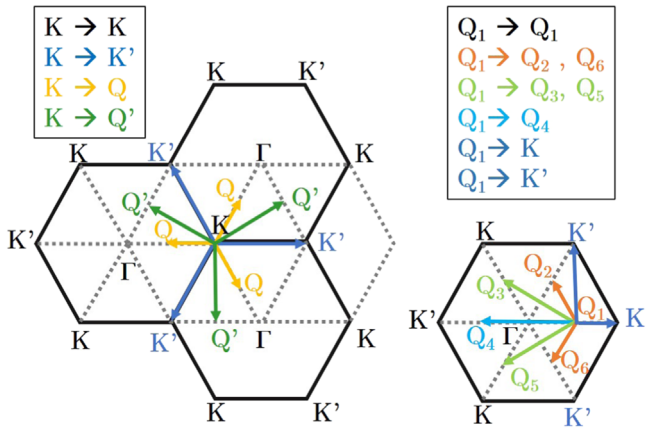

Figure 1. Possible intervalley transitions for e^-: in K valleys (left) and in Q valleys (right) of MoS_2.

2. Monte Carlo model and MoS_2 particularities

Our in-house ensemble Monte Carlo (EMC) simulator for 2D materials, which has been successfully tested for other materials such as graphene and silicene [15, 16], has been adapted to consider the specific conduction band structure and scattering properties of MoS_2. The values of the main parameters considered in the model have been extracted from the band structure defined by Li *et al* [12]. We consider a monolayer version of this hexagonal crystal system, with the conduction band minima positioned at the K points of the reciprocal lattice and the subsidiary valleys (Q) located at the intermediate point of the $\Gamma-K$ axis (70 meV over the K energy minima). The values of the longitudinal and transverse effective masses are the following: $m_K^l = m_K^t = 0.50m_0$ for the K valleys, and $m_Q^l = 0.62m_0$, $m_Q^t = 1.0m_0$ for the Q valleys, with m_0 being the electron rest mass. The phonon branches implemented in the model are the intravalley and intervalley acoustic, and the optical phonons. For the first ones, the aggregated modes involved are the transverse acoustic (TA) and the longitudinal acoustic (LA) modes; while for the second ones, transverse optical (TO(E)), longitudinal optical (LO(E)), and A_1 (homopolar) phonon modes are accounted for. According to [12, 17], they are the most relevant ones in this material, while acoustic out-of-plane modes present a weaker coupling strength and therefore they are less relevant for electronic transport. In our case, the results presented correspond to a suspended monolayer MoS_2. In the presence of a substrate, a careful treatment of the out-of-plane modes should be taken into account due to the modifications that the possible strain can generate over the structure. Indeed, there are evidences that this fact can induce changes in the vibrational and electronic properties of monolayer MoS_2 [18, 19].

The different intravalley and intervalley transitions considered for K and Q valleys are shown in figure 1. The deformation potential constants considered for electron-phonon interactions are extracted from the work of Li *et al* [12].

The ensemble Monte Carlo method is based on the simulation of a set of carriers which are subjected to the combined action of free flights under the action of the electric field, interrupted by randomly selected scattering mechanisms, with the energy-dependent scattering probabilities provided, in our case, by the deformation potential theory [12]. The simulation time is divided in equal time steps. At the end of each time step, the individual information of the dynamics of the particles is recorded and the ensemble average values of the parameters are updated. Therefore, this method allows obtaining instantaneous distribution functions and the study of transient situations. Besides the electronic transport, our Monte Carlo simulator allows us to analyse the properties at the microscopic level of a diffusive regime, which is intrinsically linked to the noise in the material under study since both the diffusive phenomenon and the noise are caused by the velocity fluctuations around its average value. The time-dependence of these velocity fluctuations is characterized by the correlation function $C_v(t)$, given by:

$$C_v(t) = <\partial v(t')\partial v(t+t')> . \tag{1}$$

where t' is the time considered once the transient is over. The value of ∂v is provided by the ensemble average over the particles simulated and obtained each time step.

The technique used to determine the diffusion coefficient D from the correlation function of velocity fluctuations has been extensively described in [15], obtaining the following relation as a function of the frequency ω:

$$D(\omega) = \int_0^\infty C_v(t)e^{j\omega t}dt, \tag{2}$$

The diffusion coefficient can be also obtained by a second technique based on the calculus of the second central moment [15]:

$$D(E) = \frac{\langle (r - \langle r \rangle)^2 \rangle}{2t}. \tag{3}$$

With E being the applied electric field, and r the parallel/perpendicular particle position, referred to the direction of the applied electric field.

Finally, the mobility μ of the material can be easily extracted from D by means of the standard Einstein relation,

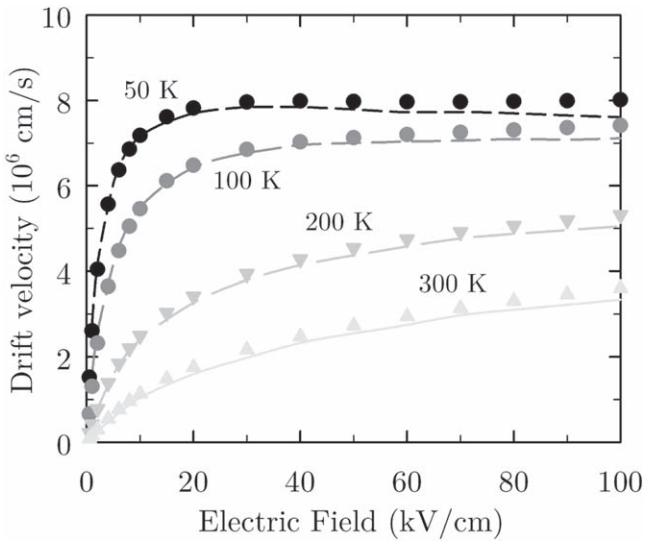

Figure 2. Drift velocity versus applied electric field for MoS_2 under different temperatures (solid lines). The results obtained by Li *et al* [12] are also shown for comparison (symbols).

since we assume non-degenerate conditions for MoS_2:

$$\mu = \frac{qD}{K_b T}, \tag{4}$$

where q is the elementary electron charge, K_b the Boltzmann constant and T the temperature.

In the simulations performed for the present work, 50 000 superparticles have been considered, with a timestep equal to 2 fs and total simulation time equal to 1000 ps.

3. Results and discussion

The dependence of the drift velocity with the applied electric field for monolayer MoS_2 under several temperatures, T, is shown in figure 2, generally presenting an increase of the carrier velocity with both the applied field and the temperature. For temperatures below or equal to 100 K it reaches a saturation value (for electric fields larger than 25 kV cm^{-1} when $T = 50$ K and 50 kV cm^{-1} when $T = 100$ K). The saturation velocity of this material at room temperature is found to be about 3.5×10^6 cm s^{-1} for E over 100 kV cm^{-1}. In comparison to graphene material [20], we observe that the drift velocity is one order of magnitude lower ($\approx 5 \times 10^7$ cm s^{-1}) [20]. Our results (solid lines) have been compared to the full-band model of Li *et al* [12] (symbols in figure 2); as it can be seen, a very good agreement is reached, therefore, highlighting the computational efficiency of our EMC model.

Our model allows us to study the individual velocity fluctuations in time by tracking any chosen particle through the whole simulation. In figure 3 we show for comparison a sample of this tracking at $T = 100$ K and $T = 300$ K at low and high electric field (1 kV cm^{-1} and 100 kV cm^{-1} respectively) for the velocity fluctuations in the parallel direction to the applied electric field. Subsequently, the perpendicular component of this magnitude is also depicted in figure 4 at the same two temperatures and low electric field (1 kV cm^{-1}). The valley in which the particle is at any moment is also included in both figures by means of a step function, showing its minimum value for the K valleys and the maximum value for the Q valleys. For $T = 100$ K and low electric field, the parallel and perpendicular components can be compared by means of figures 3(a) and 4(a). Under these low field and low temperature conditions, the particle remains at the K valleys.

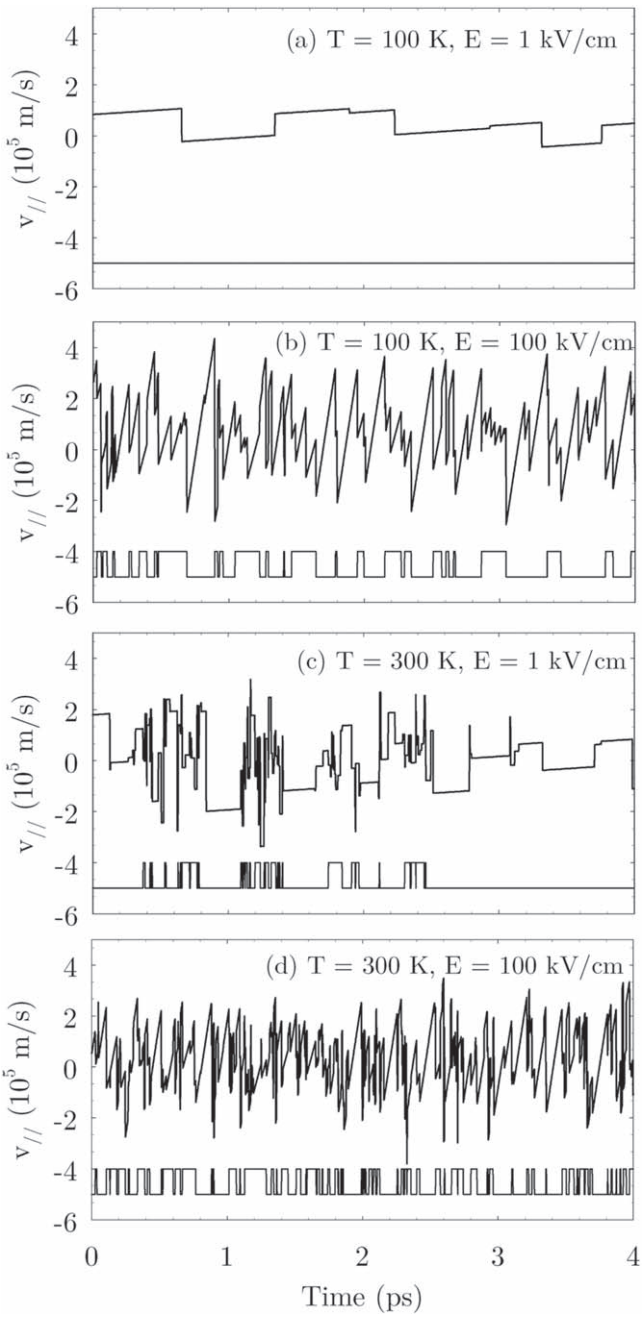

Figure 3. Instantaneous velocity in the parallel direction for a specific particle under several conditions: for $E = 1$ kV cm^{-1} and $T = 100$ K (a), for $E = 100$ kV cm^{-1} and $T = 100$ K (b), for $E = 1$ kV cm^{-1} and $T = 300$ K (c), and for $E = 100$ kV cm^{-1} and $T = 300$ K (d). In all the figures, a step function is included to depict the valley in which the particle is: minimum value for K valleys and maximum value for Q valleys.

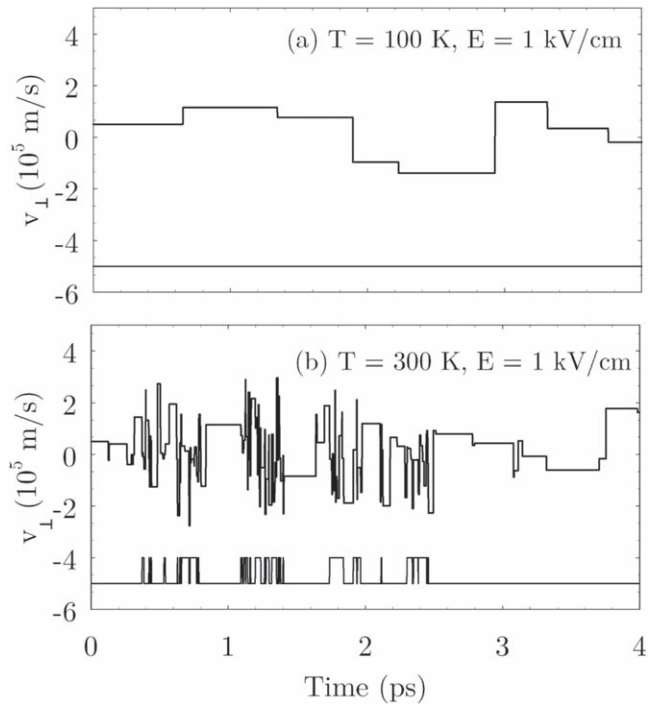

Figure 4. Instantaneous velocity in the perpendicular direction for a specific particle under several conditions: for $E = 1$ kV cm^{-1} and $T = 100$ K (a), and for $E = 1$ kV cm^{-1} and $T = 300$ K (b). In both figures, a step function is included to depict the valley in which the particle is: minimum value for K valleys and maximum value for Q valleys.

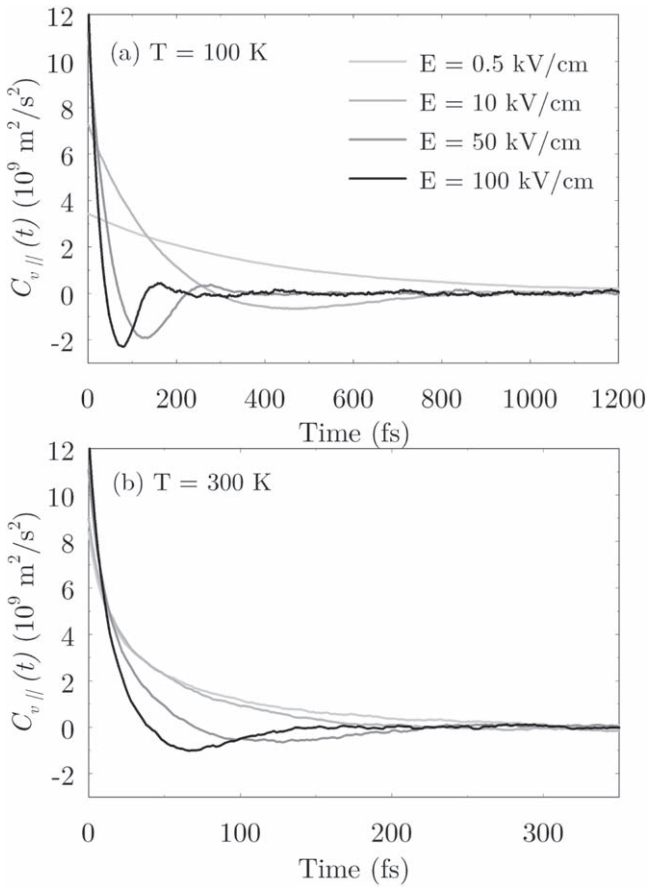

Figure 5. Autocorrelation function of parallel velocity fluctuations for several values of applied electric field and two temperatures: $T = 100$ K (a), (b), and $T = 300$ K (c), (d).

As it can be observed in both graphs, the velocity of the particle changes with time, presenting vertical oscillations from positive to negative velocity values and viceversa that are related to the re-orientation of the particle momentum after it has suffered a scattering mechanism, that in this case is of intravalley kind. The lineal evolution of carrier velocity between these oscillations corresponds to the free-flight period between scatterings, with a positive slope in the case of the parallel component, since the electric field is applied in this direction and therefore the particle increases its velocity (figure 3(a)). However, the slope for the perpendicular component of the velocity is 0 (figure 4(a)) due to the absence of an electric field component over this direction. This behaviour is analogous to the one observed in traditional semiconductors such as Silicon, but different from other 2D materials with linear dispersion, like graphene, where the existence of a limit velocity (the Fermi velocity) is distinctive. When the temperature increases, with $T = 300$ K, and under the same applied electric field (see figures 3(c) and 4(b)), the particle suffers much more scattering mechanisms as it can be learnt from the increase of fluctuations of the instantaneous velocity with time. The lineal sections are smaller in this case since the times between collisions are shorter, showing also a positive slope in the component parallel to applied field (figure 3(c)) and a flat slope in the perpendicular one (figure 4(b)). Besides, at this higher temperature the particle acquires enough energy to make intervalley transitions. As it can be seen in the step function, they are related to larger oscillations of the instantaneous velocity than in the case of intravalley transitions. The effect of increasing the applied electric field has also been analysed. In figure 3(b), the velocity fluctuations at $T = 100$ K and $E = 100$ kV cm^{-1} are shown. Under these conditions, many intervalley transitions occur, therefore wide oscillations take place accordingly. This effect is even more pronounced for larger temperatures (see figure 3(d), for $T = 300$ K and the same applied field).

The study of the velocity fluctuations shows the importance of the scattering mechanisms and the applied electric field over the electron single velocity. To evaluate more exhaustively these effects, an analysis of the ensemble autocorrelation function of these fluctuations in time is made next based on the equation (1). In figure 5 the results for the parallel components at different applied electric fields and for two temperature values are shown.

Let us focus first in the field dependence of this quantity. At low electric field $C_v(t)$ presents an exponential decay that can be approximated to the following relationship, with τ_m being the momentum relaxation time in the material [21].

$$C_v(t) = C_v(0)\exp(-t/\tau_m). \tag{5}$$

The momentum relaxation time τ_m can be estimated from the fitting of the equation (5) with the curves shown in figure 5 for low-field conditions. The obtained values range from 5×10^{-13} s at 100 K to 4×10^{-14} s at 300 K. When comparing to traditional bulk semiconductors as Silicon, it can be

noticed that MoS_2 presents shorter times (with $\tau = 1.54 \times 10^{-13}$ s y $\tau = 1.15 \times 10^{-13}$ in Si for a doping of 10^{15} cm^{-3} and 10^{17} cm^{-3} respectively [22]).

$C_v(0)$, which increases with the electric field whichever the temperature, is related to the amplitude of the velocity fluctuations, meaning that the later increase with E, as previously mentioned. On the other hand, the correlation function presents a negative maximum (related to the change of sign in the velocity fluctuations) for electric fields over 10 kV cm^{-1}, being larger as the electric field increases and shorter decay times appear, dropping to hundreds of fs at 100 kV cm^{-1}. When the temperature increases (see figure 5(b)), the negative maxima for $C_{v\|}(t)$ are smaller. This is due to a boost with the temperature of the scattering mechanisms involving the Q valleys, together with the larger corresponding effective masses in those valleys.

It can be asserted therefore that the correlation time in MoS_2 decreases with the rise of the applied electric field and/or the increase of the temperature. We will see now that it is directly related to the number of scattering mechanisms suffered by the particles. With this aim, in figure 6) the occupancy of the upper valleys and the different types of scattering mechanisms suffered by an electron in K or Q valleys are depicted as a function of the applied electric field and for several temperatures. For the sake of simplicity, we have grouped the scattering mechanisms in three different types: intravalley scattering in K-type valleys (intra-K), intravalley scattering in Q-type valleys (intra-Q) and intervalley, showing K to Q transitions which are analogous to the Q to K ones. In the case of the Q valley occupancy depicted in figure 6(a), it increases with the applied electric field. Besides, at low values of temperature and electric field, these valleys are almost unoccupied. This is consistent with the analysis of the particle tracking done in figures 3 and 4.

The influence of the intervalley transitions affects to the preponderance of the different scattering mechanisms as a function of the temperature. As it can be seen in figure 6(b), intra-K scattering mechanisms prevail at low temperatures, besides they increase in number with T, figure 6(c). Intervalley scattering mechanisms are the second more important kind of scattering mechanisms at $T = 100$ K, but they are clearly the dominant ones at $T = 300$ K. They present a boost with the temperature directly related with the increase of the average energy of the carriers and the rise of the upper valleys occupancy. Finally, the intra-Q interaction is the less important one at low temperatures due to the scarce occupancy of these valleys, figure 6(a). However, as the temperature rises also does the occupancy and the scattering mechanisms in these valleys also increase, showing a supremacy at the largest temperature studied.

Therefore, when the carriers are able to reach the Q valleys, they will be subjected to a larger number of scattering mechanisms yielding to more velocity fluctuations. This produces a larger amplitude of the fluctuations, but it is also responsible for a faster breakdown of the correlation and a larger high-frequency span of the generated noise.

A study of the diffusion coefficient can be performed from the power spectral density of the velocity fluctuations,

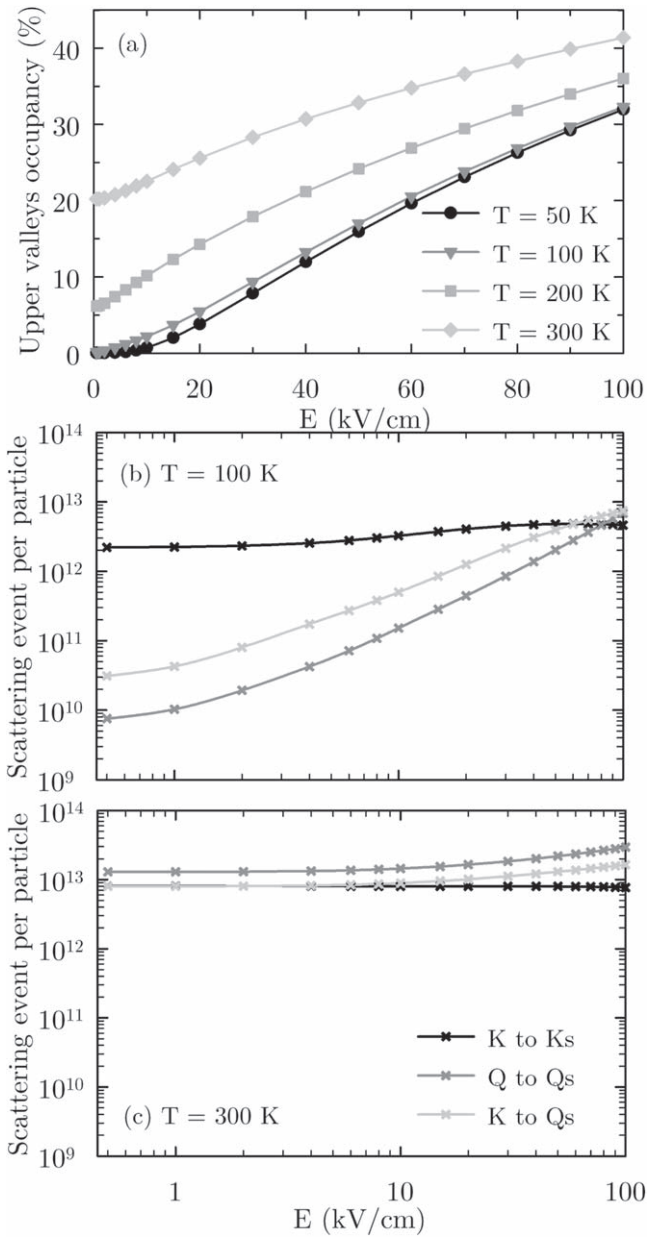

Figure 6. Occupancy of upper valleys as a function of the applied electric field for several temperatures (a). Scattering mechanism events per particle and unit of time as a function of the applied electric field at $T = 100$ K (b), and $T = 300$ K (c). Intra-K, intra-Q, intervalley K to Q transitions are shown.

that is obtained from the Fourier analysis of the correlation function [11], besides from its relation to the second central moment. We have reached coincidental results between both methodologies so we will focus on the first technique. At figure 7 the component of this quantity parallel to the applied electric field as a function of the frequency ($D(f)$) is shown for several values of the applied electric field and two temperatures. White noise behaviour in the GHz range is obtained at all the situations depicted in the figure. Focusing firstly on $T = 100$ K (see figure 7(a)), we observe that the low-frequency value of the diffusion coefficient diminishes for large electric fields and the white noise behaviour is maintained towards higher frequencies, in agreement with the shorter

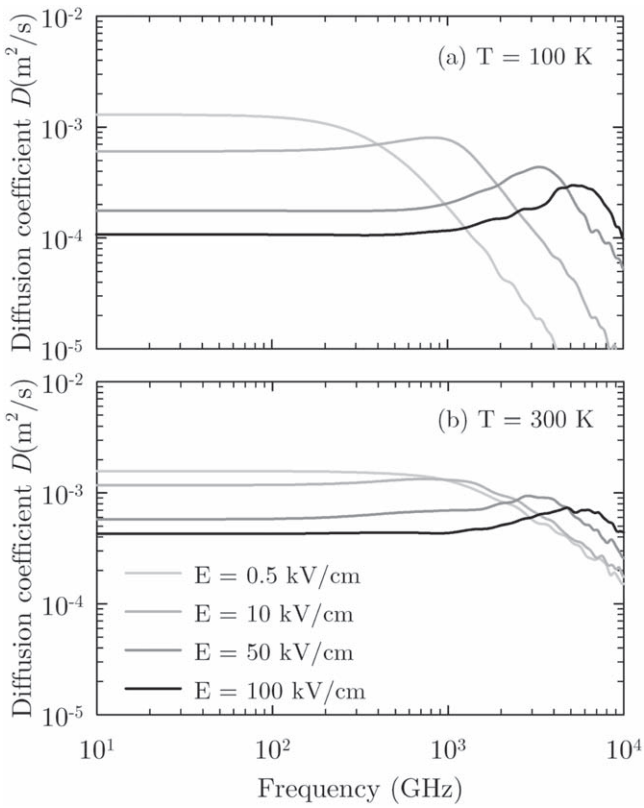

Figure 7. Diffusion coefficient for several values of applied electric field and two temperatures: $T = 100$ K (a), (b), and $T = 300$ K (c), (d).

correlation times observed in figure 5(a). Besides, a maximum connected to the negative peaks shown at $C_{v\|}(t)$ appears moving from 1 THz at $E = 10\,\mathrm{kV\,cm^{-1}}$ to 6 THz at $E = 100\,\mathrm{kV\,cm^{-1}}$. Once these maxima are surpassed, a monotonic decrease of the diffusion coefficient begins.

When the temperature rises, see figure 7(b), the same trend is depicted by $D(f)$, although the maxima are less pronounced due to the decrease of the negative maxima at $C_{v\|}(f)$ and the important increase of the intra-Q scatterings.

If we compare the values of the diffusion coefficient at $T = 300$ K, we observe that they are one order of magnitude lower than in Silicon [22] and two orders of magnitude lower than in graphene [15]. This is directly reflected in the inherent mobility of each material, as we will analyse now. Regarding the frequency dependence, the diffusion coefficient maintains the white noise behaviour until higher values in MoS_2 than in Silicon, extending up to 0.5 THz for $E = 10\,\mathrm{kV\,cm^{-1}}$ in MoS_2 and to 200 GHz in Si [22], while for the same applied electric field, in graphene the white noise spectra appears until 1 THz.

The carrier mobility in MoS_2 obtained from the diffusion coefficient is shown in figure 8. As it can be observed, the mobility decreases when both the electric field and the temperature increase, consistently with the analysis of the velocity fluctuations previously done. It is important to highlight that even for high electric fields, there exist mobility values of practical use for electronic applications, higher than $100\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$. These results are consistent with those

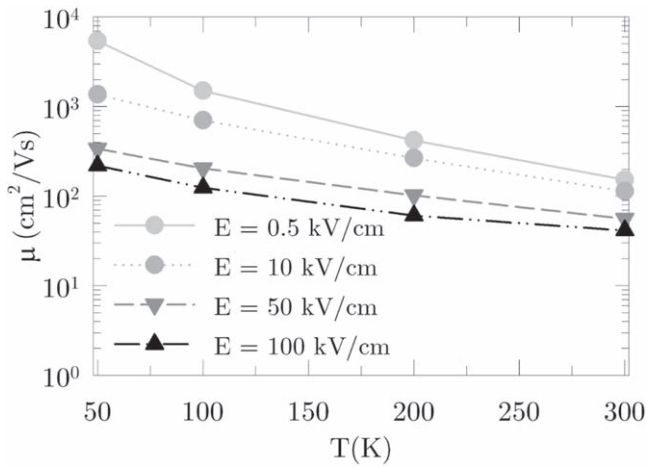

Figure 8. Carrier mobility as a function of the temperature for several applied electric fields.

determined from the slope of the velocity-field curves (figure 2) for low-field carrier mobility, being practically coincidental for temperatures larger than 100 K (particularly, at $T = 300$ K the mobility obtained from the different methodologies is approximately $150\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$). The values are very similar to those obtained by Li *et al* [12] for temperatures above 100 K. Recent experiments have yielded field effect mobility values on different MoS_2 devices ranging between 0.57 and $470\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$ [7–10]. On the other hand, when comparing the mobility of this material and that of graphene, we observe that it is at least two orders of magnitude lower than in graphene [20] and one order of magnitude lower than in Silicon, stressing the fact that the possible applications of MoS_2 will be focused not on the mobility of the material itself but on its potential device scalability and the presence of a bandgap, with great perspectives of application in such areas as foldable displays, wearable electronics or biomedicine, among others [23].

4. Conclusions

Electronic and diffusive transport properties of MoS_2 have been deeply analysed by means of an in-house ensemble Monte Carlo (EMC) simulator. Transfer to upper Q valleys presents an important dependence on the temperature and the applied field. In fact, this phenomenon, together with the scattering mechanisms suffered in the Q valleys, presents an important influence over the noise behaviour of this material. Diffusion coefficient, which is one order of magnitude lower than in Silicon and two orders of magnitude lower than in graphene, presents a reduction of its low-frequency value when the applied electric field increases, together with a wider white-noise regime. This affects to the mobility, that in the case of MoS_2 presents values ranging from 150 to $5400\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$ at low field. Therefore, it has to be taken into account that for obtaining significant mobility values in this material, low temperature conditions must be considered

since at room temperature the strong scattering activity in the upper valleys yields in a mobility deterioration.

Acknowledgments

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References

[1] Geim A K and Novoselov K S 2007 *Nat. Mater.* **6** 183
[2] Retamal J R D, Periyanagounder D, Ke J J, Tsai M L and He J H 2018 *Chem. Sci.* **9** 7727–45
[3] Mak K F, Lee C, Hone J, Shan J and Heinz T F 2010 *Phys. Rev. Lett.* **105** 136805
[4] Fivaz R and Mooser E 1967 *Phys. Rev.* **163** 743
[5] Radisavljevic B, Radenovic A, Brivio J, Giacometti I V and Kis A 2011 *Nat. Nanotechnol.* **6** 147
[6] He G *et al* 2015 *Nano Lett.* **15** 5052–8
[7] Jeong Y, Sung J Y, Choi Y, Jin J S, Yoon J H, Heo S, Hayakawa R and Wakayama Y 2019 *J. Mater. Sci.* **54** 7758–67
[8] Guros N B, Le S T, Zhang S, Sperling B A, Klauda J B, Richter C A and Balijepalli A 2019 *ACS Appl. Mater. Interfaces* **11** 16683–92
[9] Zhang S *et al* 2019 *Nanotechnology* **30** 174002
[10] Bao W, Cai X, Kim D, Sridhara K and Fuhrer M S 2013 *Appl. Phys. Lett.* **102** 042104
[11] Jacoboni C and Lugli P 1989 *The Monte Carlo Method for Semiconductor Device Simulation* (Berlin: Springer)
[12] Li X, Mullen J T, Jin Z, Borysenko K M, Buongiorno Nardelli M and Kim K W 2013 *Phys. Rev.* B **87** 115418
[13] Ferry D K 2016 *Semicond. Sci. Technol.* **31** 11LT01
[14] Nathawat J *et al* 2020 *Phys. Rev. Mater.* **4**
[15] Rengel R and Martín M J 2013 *J. Appl. Phys.* **114** 143702
[16] Rengel R, Iglesias J M, Hamham E M and Martín M J 2018 *Semicond. Sci. Technol.* **33** 065011
[17] Kaasbjerg K, Thygesen K S and Jacobsen K W 2012 *Phys. Rev.* B **85** 115317
[18] Cai Y, Lan J, Zhang G and Zhang Y W 2014 *Phys. Rev.* B **89** 035438
[19] Soni H and Jha P K 2015 *AIP Adv.* **5** 107103
[20] Feijoo P C *et al* 2019 *IEEE Trans. Electron Devices* **66** 1567–73
[21] Reggiani L, Kuhn T and Varani L 1992 *Appl. Phys.* **54** 411–27
[22] Martín M J 1996 Análisis del transporte de carga y de los fenómenos de ruido electrónico en estructuras $Si/Si_{1-x}Ge_x$ bipolares *PhD Thesis* University of Salamanca (http://nanoelec.usal.es/assets/publications/6699MariaJesusMartin-Martinez1996.pdf)
[23] Singh E, Singh P, Kim K S, Yeom G Y and Nalwa H S 2019 *ACS Appl. Mater. Interfaces* **11** 11061–105